CSE273:
Foundations of Machine Learning

UNIT-4 Part-1
(Scaler & Vectors)

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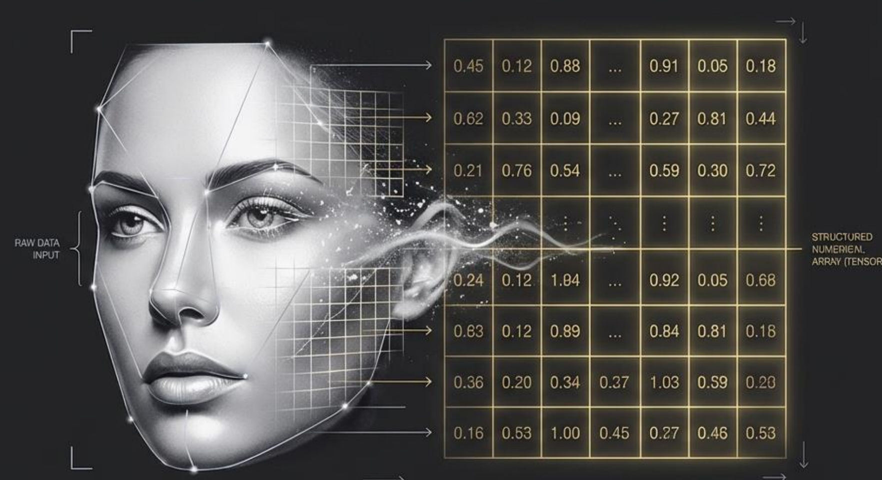
The Engine of Artificial Intelligence

Machine learning algorithms cannot process raw images, text, or audio. They process arrays of numbers. Linear algebra provides the physical rules for how these numbers are stored, structured, and transformed.



What is a Scalar?

- A scalar is a single numerical value
- It has:
  - Magnitude only
  - No direction
- Represented by lowercase letters:

$$a, b, c, \alpha$$

- Examples:
- 5, -3.2, 0, 100

Scalars in Machine Learning:

- Scalars are heavily used in ML models:
- Model parameters:
  - Weights (sometimes individual scalars)
  - Bias term
- Loss values:
  - Output of loss function is a scalar
- Learning rate:
  - Controls training speed

Scalars (0D)

Concept

A single number representing magnitude with no directional component.
The fundamental building block of all data.

Context

Used to represent individual hyperparameters (like Learning Rate), a single bias value, or the final computed Loss of a model.

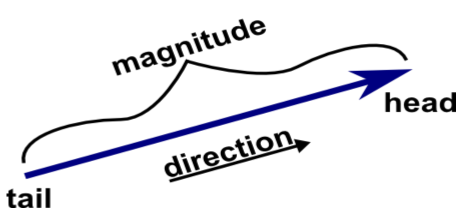
Code

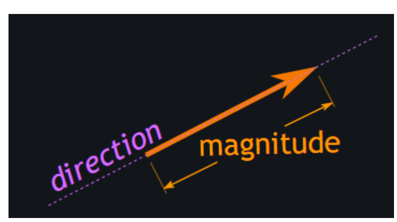
```
import numpy as np

# Creating a scalar
loss_value = np.array(0.45)
print(loss_value.shape) # Output: ()
```

What is Vector?

- Vector has magnitude and direction.
- Represented graphically as arrows.





- The length of the line shows its magnitude
- The arrowhead points in the direction.

Vectors (1D)

Concept

A one-dimensional array of numbers representing both magnitude and direction in space.

Context

Represents a single data point (a feature vector) or a sequence of learned weights in a linear regression model.

Code

```
# Creating a feature vector
features = np.array([1.5, 2.0, 3.1])
weights = np.array([0.5, -0.2, 0.1])

# Dot product (computes a scalar)
prediction = np.dot(features, weights)
```

Representation

- **Graphical Representation:** A vector is depicted as an arrow
- With a starting point (tail)
- and an ending point (head).

Vector:

[2, -6, 9]

row

or column

$$\begin{bmatrix} 2, \\ 6, \\ 9 \end{bmatrix}$$

Orthogonal Vectors:

- Vectors are orthogonal if their dot product is zero.
- Represented as $u \cdot v = 0$.
- Angle between orthogonal vectors is 90°.
- They lie perpendicular to each other.

What are the Components of a Vector?

- Vector has horizontal and vertical components.
- Horizontal component: cos θ.
- Vertical component: sin θ.

Operations:

1. Vector Addition: add corresponding components.

Formula: If $A = (a_1, a_2, ..., a_n)$

$B = (b_1, b_2, ..., b_n)$,

then $A + B = (a_1 + b_1, a_2 + b_2, ..., a_n + b_n)$.

Example: For $A = (2, 3)$ and $B = (1, 4)$, $A + B = (3, 7)$.

2. Vector Subtraction: Subtracting corresponding components.

Formula: If $A = (a_1, a_2, ..., a_n)$

$B = (b_1, b_2, ..., b_n)$,

then $A - B = (a_1 - b_1, a_2 - b_2, ..., a_n - b_n)$.

Example: For $A = (2, 3)$ and $B = (1, 4)$, $A - B = (1, -1)$.

3. Scalar Multiplication:

Multiplying each component of vector by scalar.

Formula: If $A = (a_1, a_2, ..., a_n)$

and c is a scalar,

then $cA = (c * a_1, c * a_2, ..., c * a_n)$.

Example: For $A = (2, 3)$ and $c = 2$

→ $cA = (4, 6)$.

4. Dot Product (Scalar Product)

- Dot product of two vectors results in a scalar.

Formula: If $A = (a_1, a_2, ..., a_n)$

and $B = (b_1, b_2, ..., b_n)$,

then $A \cdot B = a_1b_1 + a_2b_2 + ... + a_nb_n$.

Example: For $A = (2, 3)$ and $B = (1, 4)$

→ $A \cdot B = (2)(1) + (3)(4) = 14$.

5. Cross Product:

- Cross product results in a vector.
- Perpendicular to both original vectors.
- Defined in three-dimensional space.
- Direction determined by right-hand rule.

$$A = pi + qj + rk$$
$$B = xi + yj + zk$$

$$A \times B = \begin{vmatrix} i & j & k \\ p & q & r \\ x & y & z \end{vmatrix}$$

$$\begin{aligned} A \times B &= (qz - ry)i - (pz - rx)j + (py - qx)k \\ &= (qz - ry)i + (rx - pz)j + (py - qx)k \end{aligned}$$

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6. Magnitude (Length) of a Vector

- Length of vector from origin to its endpoint.

Formula: If $\mathbf{A} = (a_1, a_2, ..., a_n)$,

then $|\mathbf{A}| = \sqrt{(a_1^2 + a_2^2 + ... + a_n^2)}$.

Example: For $\mathbf{A} = (3, 4)$

$\rightarrow |\mathbf{A}| = \sqrt{(3^2 + 4^2)} = 5$.

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7. Unit Vector

- A unit vector has a magnitude of 1.
- Points in direction of original vector.

Formula: If $\mathbf{A} = (a_1, a_2, ..., a_n)$,

then the unit vector $\hat{\mathbf{u}} = \mathbf{A} / |\mathbf{A}|$.

Example: For $\mathbf{A} = (3, 4)$, $\hat{\mathbf{u}} = (3/5, 4/5)$.

8. Projection of a Vector

-How much of **A** is in direction of **B**.

$$\mathrm{Proj}_{\mathbf{B}}\mathbf{A} = \left(\frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{B}|^2}\right)\mathbf{B}$$

Where:

- $\mathbf{A}\cdot\mathbf{B}$ is the dot product of vectors $\mathbf{A}$ and $\mathbf{B}$,
- $|\mathbf{B}|$ is the magnitude of vector $\mathbf{B}$,
- $\mathrm{Proj}_{\mathbf{B}}\mathbf{A}$ is the projection of $\mathbf{A}$ onto $\mathbf{B}$.